A three coordinate energy bound for random permutation coordinate descent

Abstract

For every real positive definite 3×3 matrix with unit diagonal, we prove a one-epoch energy contraction bound for exact coordinate descent under a uniform random permutation. If σ is the smallest eigenvalue, the factor is $\max\{1/4, (1-\sigma)^2\}$. The proof reduces the energy deficit to three scalar polynomial inequalities and gives an exact rational Bernstein certificate over four explicitly covering boxes. The certificate has 1,595 coefficients, all nonnegative, and is regenerated directly from the six coordinate-update products. The result implies the dimension-three case of Kim–Lee–Yun's ICML 2025 conjecture and a strict asymptotic mean-square advantage over uniform sampling for almost every initialization. No assertion is made about dimensions four or higher or the optimal factor.

Keywords: random permutation coordinate descent; quadratic optimization; one-epoch contraction; mean-square convergence; Bernstein certificate

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1 Statement and source scope

Let $A \in \mathbb{R}^{3\times 3}$ be symmetric positive definite with $a_{ii} = 1$. For $i \in \{1, 2, 3\}$, set

$$P_i = I - e_i e_i^\top A.$$

This is the exact coordinate minimization map for $f(x) = \frac{1}{2}x^\top Ax$. For a permutation $\pi = (\pi_1, \pi_2, \pi_3)$, set

$$T_\pi = P_{\pi_3} P_{\pi_2} P_{\pi_1}, \qquad M = \frac{1}{6}\sum_{\pi\in S_3} T_\pi^\top A T_\pi.$$

Thus $x^\top Mx$ is the expected energy after one uniformly permuted epoch, starting from x.

Theorem 1. *For every such real matrix A, write*

$$\sigma = \lambda_{\min}(A), \qquad t = 1 - \sigma, \qquad q = \max\{1/4, t^2\}.$$

Then

$$\frac{1}{6}\sum_{\pi\in S_3} T_\pi^\top A T_\pi \preceq qA. \tag{1}$$

Kim, Lee and Yun [1] use this exact-coordinate recurrence in Section 2.2 and justify diagonal normalization there. Their Conjecture 4.1 is an asymptotic Euclidean mean-square assertion; in dimension three its proposed upper bound is

$$\beta = \max\{(2/3)^3, (1 - \sigma/3)^6\}. \tag{2}$$

Equation (1) is a stronger sufficient statement, not a quotation of the conjecture. The distinction matters: the theorem controls every initial vector after a single epoch in the A-energy, whereas the source conjecture concerns an asymptotic expected squared norm.

For a general SPD Hessian G, let $D = \operatorname{diag}(G_{11}, G_{22}, G_{33})$, $A = D^{-1/2} G D^{-1/2}$ and $y = D^{1/2} x$. Each update $x \mapsto x - e_i e_i^\top G x / G_{ii}$ becomes $y \mapsto P_i y$. Hence the theorem applies with $\sigma = \lambda_{\min}(D^{-1/2} G D^{-1/2})$, and its energy is exactly $x^\top G x$. A translated quadratic is handled by subtracting its minimizer.

2 A domain that includes every real sign pattern

The trace of A is three, so $0 < \sigma \le 1$ and $0 \le t < 1$. If $t = 0$, then $A = I$ and $T_\pi = 0$, proving the theorem. Suppose henceforth that $t > 0$ and write

$$A = \begin{pmatrix} 1 & ta & tb \\ ta & 1 & tc \\ tb & tc & 1 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & a & b \\ a & 1 & c \\ b & c & 1 \end{pmatrix}. \tag{3}$$

Since $A - \sigma I = tC$, the matrix C is positive semidefinite and singular. Its principal minors and determinant give

$$|a|, |b|, |c| \le 1, \qquad a^2 + b^2 + c^2 = 1 + 2abc. \tag{4}$$

Introduce the symmetric invariants

$$p = abc, \qquad r = a^2 b^2 + a^2 c^2 + b^2 c^2. \tag{5}$$

Lemma 1. *Every triple in* (4) *satisfies* $-1/8 \le p \le 1$ *and*

$$p \ge 0: \quad 2p + p^2 \le r \le \frac{(1+2p)^2}{3}, \tag{6}$$

$$p \le 0: \quad 0 \le r \le \frac{1}{3}. \tag{7}$$

Proof. The upper bound $p \le 1$ follows from $|a|, |b|, |c| \le 1$. For $p < 0$, put $z = (-p)^{1/3} \ge 0$. The arithmetic–geometric mean inequality gives $1 - 2z^3 = a^2 + b^2 + c^2 \ge 3z^2$. Since

$$2z^3 + 3z^2 - 1 = (2z - 1)(z + 1)^2,$$

we have $z \le 1/2$, proving $p \ge -1/8$. Furthermore,

$$0 \le (1 - a^2)(1 - b^2)(1 - c^2) = r - 2p - p^2,$$

and $r \le (a^2 + b^2 + c^2)^2/3 = (1 + 2p)^2/3$. These imply (6); for $p \le 0$ the latter upper bound is at most $1/3$, while $r \ge 0$ directly. ∎

These are outer bounds on the invariant domain. It is unnecessary that every point satisfying them comes from a correlation matrix. Proving positivity on these larger sets covers all admissible matrices, including mixed signs and zero off-diagonal entries.

3 An exact energy deficit and its real spectrum

Define

$$H = A^{-1}(A - M)A^{-1}, \qquad H_6 = 6H. \tag{8}$$

Direct multiplication of the six coordinate products gives

$$\begin{aligned}
(H_6)_{11} &= 6 + 3t^2(a^2 + b^2) - 4t^3abc + t^4c^2(a^2 + b^2),\\
(H_6)_{22} &= 6 + 3t^2(a^2 + c^2) - 4t^3abc + t^4b^2(a^2 + c^2),\\
(H_6)_{33} &= 6 + 3t^2(b^2 + c^2) - 4t^3abc + t^4a^2(b^2 + c^2),\\
(H_6)_{12} &= -6ta + 4t^2bc - t^3a(b^2 + c^2),\\
(H_6)_{13} &= -6tb + 4t^2ac - t^3b(a^2 + c^2),\\
(H_6)_{23} &= -6tc + 4t^2ab - t^3c(a^2 + b^2).
\end{aligned} \tag{9}$$

The matrix is symmetric. An especially direct verification of the formula, avoiding a symbolic inverse, is the polynomial identity

$$AH_6A = 6A - \sum_{\pi\in S_3} T_\pi^\top AT_\pi. \tag{10}$$

This identity holds for arbitrary real a, b, c, t, without imposing (4), and is checked exactly in the supplied verifier.

Define the polynomial matrix

$$V = \frac{AH_6 - 6(1 - t^2)I}{t^2}. \tag{11}$$

Each numerator entry is divisible by t^2, so V extends polynomially to $t = 0$. Although V need not be symmetric, it is similar to the symmetric matrix

$$A^{-1/2}VA^{1/2} = \frac{A^{1/2}H_6A^{1/2} - 6(1 - t^2)I}{t^2}. \tag{12}$$

All eigenvalues of V are therefore real. Congruence shows that $M \preceq t^2A$ is equivalent to their nonnegativity. For the remaining branch $q = 1/4$, use

$$W = (3/2 - 6t^2)I + t^2V = AH_6 - \tfrac{9}{2}I. \tag{13}$$

Again W is similar to a real symmetric matrix; its nonnegative spectrum is equivalent to $M \preceq A/4$.

For a 3×3 matrix Z, let

$$e_1(Z) = \operatorname{tr} Z, \quad e_2(Z) = \tfrac{1}{2}((\operatorname{tr} Z)^2 - \operatorname{tr}(Z^2)), \quad e_3(Z) = \det Z.$$

Lemma 2. *If Z has real eigenvalues and $e_1(Z), e_2(Z), e_3(Z) \geq 0$, then all its eigenvalues are nonnegative.*

Proof. For $s > 0$, the polynomial $\det(sI + Z) = s^3 + e_1(Z)s^2 + e_2(Z)s + e_3(Z)$ is strictly positive. A negative eigenvalue $-s$ would make it zero. ∎

The real-spectrum hypothesis in this step is supplied by (12), rather than inferred from the signs of the polynomial coefficients.

4 The exact scalar certificate

Expanding $e_j(V)$ and using (4) and (5) yields polynomials I_1, I_2, I_3 in t, p, r. The complete formulas are specified by (9)–(11) and the elementary-invariant definitions; Appendix A gives their scalar coefficient form as well. In particular,

$$I_1 = 12 + 12p(t-1) - 2rt^2, \qquad I_3 = (1-t)F(t,p,r), \tag{14}$$

where F is a polynomial. Put $\alpha = 3/2 - 6t^2$. The invariants of W are

$$\begin{aligned} J_1 &= 3\alpha + t^2 I_1, \\ J_2 &= 3\alpha^2 + 2\alpha t^2 I_1 + t^4 I_2, \\ J_3 &= \alpha^3 + \alpha^2 t^2 I_1 + \alpha t^4 I_2 + t^6 I_3. \end{aligned} \tag{15}$$

All four cases below use $(\rho, \tau, \eta) \in [0,1]^3$:

Case	t	p	r
Large t, nonnegative p	$(1+\tau)/2$	ρ	$2\rho + \rho^2 + (1-\rho)^2\eta/3$
Large t, nonpositive p	$(1+\tau)/2$	$-\rho/8$	$\eta/3$
Small t, nonnegative p	$\tau/2$	ρ	$2\rho + \rho^2 + (1-\rho)^2\eta/3$
Small t, nonpositive p	$\tau/2$	$-\rho/8$	$\eta/3$

Lemma 1 proves coverage. For $p \geq 0$, the width of the r interval is $(1-p)^2/3$. At $p = 1$ that interval collapses to $r = 3$ and every η has the same image, so this boundary is also included. Both branches include $t = 1/2$ and both sign cases include $p = 0$.

For a polynomial $f(z) = \sum_k c_k z^k$ with coordinate degrees $d = (d_1, d_2, d_3)$, define

$$b_i = \sum_{k \leq i} c_k \prod_{j=1}^{3} \frac{\binom{i_j}{k_j}}{\binom{d_j}{k_j}}, \qquad 0 \leq i_j \leq d_j. \tag{16}$$

The inequalities $k \leq i$ are coordinatewise. Its tensor Bernstein expansion is exactly

$$f(z) = \sum_{0 \leq i \leq d} b_i \prod_{j=1}^{3} \binom{d_j}{i_j} z_j^{i_j} (1-z_j)^{d_j - i_j}. \tag{17}$$

To verify (16), apply the univariate identity $z^k = \sum_{i=k}^{d} \binom{i}{k}\binom{d}{k}^{-1}\binom{d}{i} z^i (1-z)^{d-i}$ in each variable. Every basis term in (17) is nonnegative on the unit cube. Consequently $b_i \geq 0$ for every index is an exact positivity certificate; no grid sampling, limit argument or floating-point tolerance enters it.

Table 1 records the degree, number and minimum of the coefficients computed by (16). The coefficient ordering is lexicographic in (i_ρ, i_τ, i_η), with i_η varying fastest. The polynomials, substitutions, and formula (16) determine every coefficient uniquely. The accompanying JSON lists all 1,595 rational numbers, and the standalone verifier both regenerates

Case and polynomial	Degree (ρ, τ, η)	Count	Minimum	Zeros
Large t, $p \geq 0$: I_1	$(2, 2, 1)$	18	9/2	0
I_2	$(4, 6, 2)$	105	0	3
F	$(6, 8, 2)$	189	0	3
Large t, $p \leq 0$: I_1	$(1, 2, 1)$	12	34/3	0
I_2	$(4, 6, 2)$	105	295915/9216	0
F	$(5, 8, 2)$	162	81	0
Small t, $p \geq 0$: J_1	$(2, 4, 1)$	30	9/8	0
J_2	$(4, 10, 2)$	165	81/256	0
J_3	$(6, 15, 2)$	336	27/1024	0
Small t, $p \leq 0$: J_1	$(1, 4, 1)$	20	71/24	0
J_2	$(4, 10, 2)$	165	171/64	0
J_3	$(5, 15, 2)$	288	81/128	0

Table 1: Exact rational Bernstein certificate. The six zero coefficients occur only on the large-t, nonnegative-p branch; nonnegativity is sufficient.

them from the six updates and re-expands (17) to check each polynomial identity independently.

Proof of Theorem 1. The case $t = 0$ was established above. For $0 < t \leq 1/2$, the two small-t rows of the domain decomposition cover all admissible p, r. The nonnegative Bernstein coefficients prove $J_1, J_2, J_3 \geq 0$. Lemma 2, the real spectrum of W, and (13) give $M \preceq A/4$. For $1/2 \leq t < 1$, the other two boxes prove $I_1, I_2, F \geq 0$. Since $1 - t > 0$, equation (14) implies $I_3 \geq 0$. Apply Lemma 2 to V and use (12) to obtain $M \preceq t^2 A$. These cases give (1). ∎

The certificates include $t = 1$ as a closed-box endpoint, but the theorem itself assumes positive definiteness and therefore $t < 1$. No inversion of a singular A is used at that endpoint.

5 Asymptotic comparison with uniform sampling

Let x_k^{RP} be the RPCD output after k independent uniform epochs. Conditional expectation and Theorem 1 give

$$\mathbb{E}\|x_k^{\mathrm{RP}}\|_A^2 \leq q^k \|x_0\|_A^2, \qquad \mathbb{E}\|x_k^{\mathrm{RP}}\|^2 \leq \kappa(A) q^k \|x_0\|^2,$$

where $\kappa(A) = \lambda_{\max}(A)/\lambda_{\min}(A)$. In particular the asymptotic upper rate is at most q in squared Euclidean norm.

A self-contained lower comparison for RCD is short. Let y_m be its output after m independent uniform coordinate updates. A coordinate i is untouched with probability $(2/3)^m$, and on that event $(y_m)_i = (x_0)_i$. Consequently

$$\mathbb{E}\|y_m\|^2 \geq (x_0)_i^2 (2/3)^m. \tag{18}$$

Also $\mathbb{E}y_m = (I - A/3)^m x_0$. Let Π_σ be the orthogonal projection onto the σ-eigenspace of A. Jensen's inequality yields

$$\mathbb{E}\|y_m\|^2 \geq \|\mathbb{E}y_m\|^2 \geq (1 - \sigma/3)^{2m}\|\Pi_\sigma x_0\|^2. \tag{19}$$

Outside the proper subspace $\ker \Pi_\sigma$, equations (18)–(19) give a positive constant $c(A, x_0)$ such that

$$\mathbb{E}\|y_{3k}\|^2 \geq c(A, x_0)\beta^k,$$

with β as in (2). That excluded subspace has Lebesgue measure zero. If the first term defines β, equation (18) alone suffices for every nonzero initialization.

Finally $1/4 < 8/27$. For $0 \leq t < 1$, the arithmetic–geometric mean inequality applied to $1, 1, t$ gives

$$t^2 < \left(\frac{2+t}{3}\right)^6 = (1 - \sigma/3)^6.$$

The inequality is strict because equality would require $t = 1$ (and the case $t = 0$ is immediate). Therefore $q < \beta$ for every SPD instance.

Corollary 5.1. *For every real SPD three-coordinate quadratic, the RPCD asymptotic squared-norm upper rate is strictly smaller than the RCD lower rate per three updates, for almost every initialization. More strongly, for every initialization outside the exceptional subspace above,*

$$\frac{\mathbb{E}\|x_k^{\mathrm{RP}}\|^2}{\mathbb{E}\|y_{3k}\|^2} \longrightarrow 0$$

at a geometric rate bounded by $q/\beta < 1$. In particular Theorem 1 implies the dimension-three upper bound in Conjecture 4.1 of [1].

This comparison uses a limsup for the RPCD rate and a liminf for the RCD rate, so it does not need a separate existence theorem for either rate limit. If those limits are used, the same inequalities hold. Fixed diagonal coordinate scaling changes norm comparison constants only, so the conclusion also holds for non-unit diagonals.

A Explicit invariant polynomials

Here are compact scalar formulas sufficient to regenerate the certificate without symbolic matrix multiplication. Let

$$I_2 = \sum_{k=0}^{6} U_k(p, r)t^k, \qquad F = \sum_{k=0}^{8} D_k(p, r)t^k.$$

The coefficients below, together with (14) and (15), determine all twelve transformed polynomials in Table 1.

$$
\begin{aligned}
U_0 &= 36p^2 - 108p + 5r + 45 \\
U_1 &= -32p^2 + 128p \\
U_2 &= -72p^3 - 99p^2 + 20pr - 54p - 14r - 9 \\
U_3 &= 136p^3 + 136p^2 - 14pr + 34p \\
U_4 &= -80p^3 - 12p^2r - 40p^2 - 12pr + r^2 - 3r \\
U_5 &= 18p^3 + 12p^2r + 6pr \\
U_6 &= -4p^4 - 4p^3 - p^2r - p^2
\end{aligned}
$$

$$
\begin{aligned}
D_0 &= 191p^2 - 30pr - 216p + 15r + 54 \\
D_1 &= -p^2 - 30pr + 120p + 15r + 54 \\
D_2 &= -366p^3 + 84p^2r - 562p^2 + 174pr - 204p - 10r^2 + 24r \\
D_3 &= 410p^3 - 28p^2r + 254p^2 + 34pr - 10r^2 + 24r \\
D_4 &= -24p^4 - 48p^3r - 94p^3 - 142p^2r + 8p^2 + 12pr^2 - 74pr + 2r^2 \\
D_5 &= 4p^4 + 104p^3r + 28p^3 + 82p^2r + 8p^2 + 2r^2 \\
D_6 &= -16p^5 - 25p^4 - 46p^3r - 8p^3 - 8p^2r^2 + p^2r - 8pr^2 \\
D_7 &= 16p^5 + 7p^4 + 2p^3r + 8p^2r^2 + p^2r \\
D_8 &= 2p^5 - 4p^4r - 2p^3r
\end{aligned}
$$

B Exact verification and provenance

The supplementary files are `verify_bernstein_certificate.py` and `bernstein_certificate.json`. The verifier requires Python 3 and SymPy and uses exact integers and rationals throughout. Running it without arguments checks the saved certificate against a fresh computation; `--write` regenerates the same certificate.

Its verification chain is explicit: construct the three P_i; enumerate all six products in temporal order; check (10); form V; compute its elementary invariants; reduce the symmetric polynomials in a^2, b^2, c^2 using their sum $1 + 2p$, pair sum r, and product p^2; verify the factor $1 - t$ in the determinant; form the shifted invariants (15); apply the four substitutions; compute every rational coefficient by (16); check its sign; and reconstruct the polynomial from the Bernstein basis. An additional separate script recomputes H by symbolic inversion from the six products and confirms the same three invariant polynomials. These are exact algebraic checks, not numerical semidefinite evidence.

The JSON certificate has SHA-256

`ec08856e5cda862fa4033cb6c2d26af7e0a7`
`d93c70388aa46e39cd5c23550f2b.`

The digest identifies this new certificate; mathematical validity rests on the identities, domain coverage and coefficient signs above. The previously reported 108-coefficient table was unavailable and was neither reconstructed by guessing nor relied on. This note and the 1,595-coefficient certificate are newly derived artifacts, with no claim of byte identity to the lost manuscript.

The primary PDF was retrieved from the official PMLR proceedings link on October 1, 2026. Its SHA-256 is

`05ebc38113d74805378838c5a3635e4c1e7`
`5756c5778ea9ba3ec79a215ab1ea2.`

This source digest is distinct from any historical manuscript digest.

References

[1] Donghwa Kim, Jaewook Lee and Chulhee Yun. *Provable Benefit of Random Permutations over Uniform Sampling in Stochastic Coordinate Descent*. Proceedings of the 42nd International Conference on Machine Learning, PMLR 267:30480–30547, 2025. `https://proceedings.mlr.press/v267/kim25x.html`.